PACKET TRACER PROJECT

Enterprise Multi-Site Incident Response

Network design, configuration baseline and troubleshooting of five incident tickets

20 devices | 4 regions plus HQ | OSPF, EIGRP, HSRPv2, ROAS, DHCP, ACLs

October 2026

Table of Contents

1. Executive Summary 3

2. Network Topology & Architecture 4

2.1 Topology overview and site roles 4
2.2 IP addressing plan and VLAN design 4
2.3 Routing design 5
2.4 Layer 2 design and redundancy 5
2.5 Network services 5

3. Configuration Documentation 6

3.1 Baseline and management access 6
3.2 WAN and core configuration 6
3.3 Site routers 7
3.4 Switching configuration 8
3.5 ACL policy 9
3.6 Baseline verification 11

4. Incident Response & Ticket Resolution 13

4.1 Methodology 13
4.2 TK001 – HSRP version mismatch 13
4.3 TK002 – Router-on-a-stick failure 14
4.4 TK003 – Route redistribution failure 14
4.5 TK004 – DHCP snooping trust 15
4.6 TK005 – ACL faults 15
4.7 Summary matrix 16

5. Key Takeaways & Lessons Learned 17

5.1 Troubleshooting methodology 17
5.2 Recurring fault patterns 17
5.3 Preventive practices 17
5.4 Lab and tool observations 17

6. Conclusion & Future Enhancements 18

6.1 Conclusion 18
6.2 Project limitations 18
6.3 Future enhancements 18

7. Appendices 19

Appendix A – Inventory, links and addressing 19
Appendix B – Running configurations 21
Appendix C – Command quick reference 21
Appendix D – Individual ticket reports 22

1. Executive Summary

This document records the design, configuration and troubleshooting of a multi-site enterprise network built in Cisco Packet Tracer. A two-router headquarters (HQ) core connects four regional sites (Regions A to D) through dedicated Edge routers. The network contains 20 devices: 10 Cisco 2911 routers, 2 Catalyst 3650 distribution switches and 8 Catalyst 2960 access switches.

OSPF area 0 forms the backbone, and each region runs its own EIGRP autonomous system (AS 1 to 4). The Edge routers redistribute between the two protocols. Region D uses a pair of distribution switches with HSRPv2 for gateway redundancy, and the other regions use router-on-a-stick. Supporting services include DHCP with DHCP snooping and Dynamic ARP Inspection, port security, DNS, syslog and NTP. Management is by SSH only, restricted by ACL to the Network Engineer workstation and to each region's technician. Simple, practical ACLs on the HQ and Edge routers control traffic between sites.

At a glance	
Devices	20 (10 routers, 2 distribution switches, 8 access switches)
Sites	HQ and four regions (A to D)
Routing	OSPF area 0 backbone, EIGRP AS 1 to 4 per region, mutual redistribution on Edge-R1 to Edge-R4
Redundancy	Dual serial links between HQ routers, HSRPv2 in Region D, LACP Port-channel between the distribution switches
Incidents handled	5 tickets (TK001 to TK005), 7 individual faults
Method	Layer 1 to Layer 4 troubleshooting, one layer ruled out at a time

After the network was configured and fully verified, five incidents were introduced and resolved:

- **TK001:** HSRP version mismatch between DSW1 and DSW2
- **TK002:** router-on-a-stick failure caused by a VLAN missing from a trunk
- **TK003:** missing route redistribution on Edge-R2
- **TK004:** DHCP snooping trust missing on an uplink
- **TK005:** three ACL faults (wrong interface direction, wrong port position and wrong address match)

In most cases interfaces were up and routing neighbors were established, so the faults were found by checking one layer at a time rather than by looking at link status. Each ticket is documented with its symptoms, evidence, root cause, fix and verification, and the results feed a set of lessons on troubleshooting method, configuration consistency and change verification.

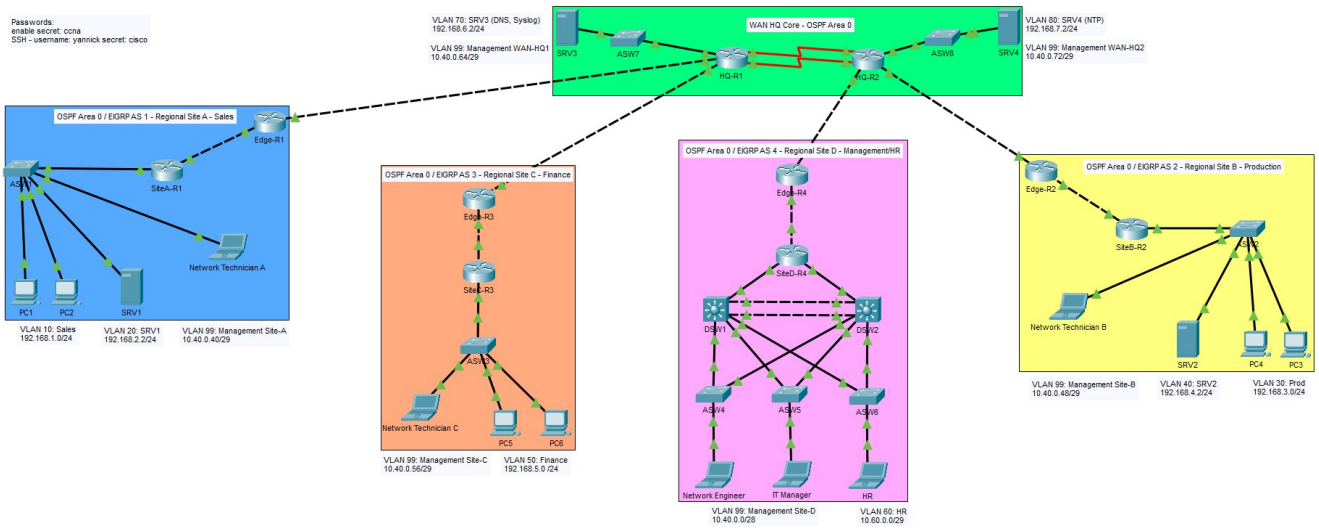
How this document is organized. Section 2 describes the topology and design. Section 3 documents the configuration of every device group and the healthy baseline. Section 4 covers the five tickets. Sections 5 and 6 give the lessons learned, limitations and future enhancements. The appendices contain the inventory and addressing tables, a command reference and the individual ticket reports.

2. Network Topology & Architecture

2.1 Topology overview and site roles

The network is a hub-and-spoke design. HQ-R1 and HQ-R2 form the core. Each region connects to the core through one Edge router: Edge-R1 and Edge-R3 attach to HQ-R1, and Edge-R2 and Edge-R4 attach to HQ-R2.

Site	WAN edge	Site router and switching	Role
HQ	HQ-R1, HQ-R2 (two serial links between them)	ASW7 (SRV3), ASW8 (SRV4)	Core and shared services
Region A	Edge-R1 (via HQ-R1)	SiteA-R1, ASW1	Sales users, SRV1
Region B	Edge-R2 (via HQ-R2)	SiteB-R2, ASW2	Production users, SRV2
Region C	Edge-R3 (via HQ-R1)	SiteC-R3, ASW3	Finance users
Region D	Edge-R4 (via HQ-R2)	SiteD-R4, DSW1, DSW2, ASW4 to ASW6	HR, IT Manager and Network Engineer



Three design choices shape the rest of the document:

- **One boundary router per region.** Each Edge router is the only point where OSPF and EIGRP meet, so redistribution cannot create routing loops.
- **Redundancy where it matters.** HQ has two routers and two serial links (primary and backup), and Region D has two distribution switches. Every other region has a single WAN path.
- **A dedicated management VLAN (VLAN 99)** at every site, so that device management stays separate from user traffic.

2.2 IP addressing plan and VLAN design

Point-to-point links use /30 subnets and management VLANs use /29 or /28 subnets (VLSM), which keeps address use efficient. The complete tables are in Appendix A.

Range	Use
10.0.0.0/30, 10.0.0.4/30	HQ-R1 to HQ-R2 serial links (primary and backup)
10.10.0.0/30, 10.20.0.0/30, 10.30.0.0/30	HQ to Edge-R1, Edge-R2 and Edge-R3
10.10.0.4/30, 10.20.0.4/30, 10.30.0.4/30	Edge router to site router, Regions A, B and C
10.40.0.x /30	Region D transit and inter-switch links, including HQ-R2 to Edge-R4

Range	Use
10.40.0.0/28 and 10.40.0.40, .48, .56, .64, .72 /29	Management VLAN 99 (Region D, Regions A to C, HQ)
192.168.1.0 to 192.168.5.0 /24	User and server VLANs, Regions A to C
192.168.6.0/24, 192.168.7.0/24	HQ server VLANs (SRV3 and SRV4)
10.60.0.0/29	Region D HR VLAN
1.1.1.1 to 13.13.13.13 /32	One loopback per router and distribution switch

VLAN	Name	Subnet	Default gateway
10 / 20	Sales / SRV1 (Region A)	192.168.1.0/24, 192.168.2.0/24	SiteA-R1 G0/0.10 and G0/0.20
30 / 40	Prod / SRV2 (Region B)	192.168.3.0/24, 192.168.4.0/24	SiteB-R2 G0/0.30 and G0/0.40
50	Finance (Region C)	192.168.5.0/24	SiteC-R3 G0/0.50
60	HR (Region D)	10.60.0.0/29	HSRP virtual IP 10.60.0.1
70 / 80	SRV3 / SRV4 (HQ)	192.168.6.0/24, 192.168.7.0/24	HQ-R1 G0/2.70 and HQ-R2 G0/0.80
99	Management (all sites)	See Appendix A.5	Site router, or HSRP virtual IP 10.40.0.1 in Region D
999	Native (unused)	–	–

2.3 Routing design

- **Backbone:** OSPF process 1, area 0, runs on HQ-R1, HQ-R2 and the four Edge routers (WAN-facing interfaces, loopbacks and the HQ server and management networks).
- **Regions:** each region runs its own EIGRP AS: AS 1 (A), AS 2 (B), AS 3 (C) and AS 4 (D, which includes the distribution switches).
- **Boundary:** every Edge router is an ASBR (Autonomous System Boundary Router) that redistributes in both directions, as shown below.
- **WAN cost:** the primary serial link uses the default OSPF cost (64) and the backup link has `ip ospf cost 200`. Both adjacencies stay FULL, and the backup carries traffic only if the primary fails.

```
router eigrp <AS>
 redistribute ospf 1 metric 1000000 100 255 1 1500
router ospf 1
 redistribute eigrp <AS> subnets
```

EIGRP needs the metric (bandwidth in kbps, delay, reliability, load, MTU) because it has no default seed metric. OSPF assigns a default seed metric of 20 to redistributed routes (type E2), and `subnets` is needed so that non-classful subnets are included.

2.4 Layer 2 design and redundancy

- **Router-on-a-stick** serves Regions A, B and C and the two HQ server VLANs, over 802.1Q trunks with native VLAN 999.
- **Region D** uses a Layer 3 distribution pair. DSW1 and DSW2 are routed to SiteD-R4 and are joined by a routed Port-channel 1 (LACP).
- **HSRPv2:** group 1 (VLAN 60) and group 2 (VLAN 99). DSW1 is active for VLAN 99 and DSW2 for VLAN 60, each with priority 105 and preemption.
- **Spanning tree:** Rapid-PVST, with the root placed on the same switch as the HSRP active router for each VLAN.
- **HSRP hello path:** because Port-channel 1 is routed, HSRP hellos between the DSWs cross the access layer, through ASW4. This matters in TK001.
- **WAN redundancy** exists only at HQ (dual serial links). Each region has a single WAN path.

2.5 Network services

Service	Provider	Notes
DHCP	SiteA-R1, SiteB-R2, SiteC-R3	One pool per user VLAN, addresses .1 to .10 excluded. Region D and HQ use static addresses.
DNS	SRV3 (192.168.6.2)	Configured on every device and handed out by DHCP
Syslog	SRV3 (192.168.6.2)	UDP 514, trap level debugging, from every device
NTP	SRV4 (192.168.7.2)	Every device synchronizes to SRV4
Web	SRV1 (Region A), SRV2 (Region B)	Regional web servers used in TK002 and TK005
Management	SSH version 2 only	Network Engineer everywhere, technicians in their own region (Section 3.1)
Switch security	ASW1 to ASW8	Port security everywhere. DHCP snooping and DAI on ASW1 to ASW3.

3. Configuration Documentation

This section documents what is configured on each device group and why. Full running configurations are directly available in the Packet Tracer file. The section ends with the baseline of the healthy network, which the ticket reports in Section 4 compare against.

3.1 Baseline and management access

The following baseline applies to all 20 devices:

Area	Configuration
Naming and DNS	Hostname, domain `yannickccna.com`, name-server 192.168.6.2
Authentication	`enable secret`; local user `yannick`; VTY lines use `login local`; the console has no login by design
Remote access	SSH version 2 only (`transport input ssh`) on VTY lines 0 to 15
Access control	Standard ACL `SSH-Restriction` applied with `access-class … in` on every VTY line
Console	`exec-timeout 45 0`, `logging synchronous`
Logging	`logging 192.168.6.2` (SRV3), trap level `debugging`
Time	`ntp server 192.168.7.2` (SRV4)
Discovery	LLDP enabled on devices and disabled on access ports
Housekeeping	Unused interfaces and Vlan1 shut down

SSH is allowed from three kinds of source, enforced by `SSH-Restriction` on each device:

Source	Devices it may SSH to
Network Engineer (10.40.0.4)	All 20 devices
Technician A (10.40.0.42)	SiteA-R1, ASW1, Edge-R1 (via its loopback)
Technician B (10.40.0.50)	SiteB-R2, ASW2, Edge-R2 (via its loopback)
Technician C (10.40.0.58)	SiteC-R3, ASW3, Edge-R3 (via its loopback)

HQ, Region D and Edge-R4 devices accept only the Network Engineer. The Edge routers need a second permit, because their ACLs must also let the technician's SSH traffic through. If SSH is unreachable because of a fault, the console is the fallback.

3.2 WAN and core configuration

3.2.1 HQ-R1 – core router 1

HQ-R1 connects Edge-R1 (Region A) and Edge-R3 (Region C) to the core. It also connects to HQ-R2 over two serial links and hosts the SRV3 VLAN and the HQ management VLAN through ASW7.

Function	Configuration	Purpose
WAN to Edge routers	G0/0 `10.10.0.1/30`, G0/1 `10.30.0.1/30`, `ip ospf network point-to-point`	No DR/BDR election, faster adjacency
Core links to HQ-R2	S0/3/0 `10.0.0.1/30` (primary), S0/3/1 `10.0.0.5/30` (backup, `ip ospf cost 200`), `clock rate 4000000`	Primary preferred, backup on failure. HQ-R1 is the DCE side.
Router-on-a-stick	G0/2.70 `192.168.6.1/24` (SRV3), G0/2.99 `10.40.0.65/29` (management)	One physical link to ASW7 carries both VLANs
Routing	OSPF 1, router-id 1.1.1.1, all networks in area 0	Backbone participation
Passive interfaces	Loopback0, G0/2, G0/2.70, G0/2.99	No OSPF hellos towards hosts or switches
Traffic control	`SRV3-Inbound` applied **out** on G0/2.70 and G0/2.99	See Section 3.5

3.2.2 HQ-R2 – core router 2

HQ-R2 mirrors HQ-R1. It connects Edge-R2 (Region B) and Edge-R4 (Region D) and hosts the SRV4 VLAN and the HQ management VLAN through ASW8.

Function	Configuration	Difference from HQ-R1
WAN to Edge routers	G0/1 `10.20.0.1/30`, G0/2 `10.40.0.33/30`, both OSPF point-to-point	Edge-R2 and Edge-R4 instead of Edge-R1 and Edge-R3
Core links	S0/3/0 `10.0.0.2/30` (primary), S0/3/1 `10.0.0.6/30` (backup, cost 200)	DTE side: no `clock rate`
Router-on-a-stick	G0/0.80 `192.168.7.1/24` (SRV4), G0/0.99 `10.40.0.73/29` (management)	VLAN 80 instead of VLAN 70
Routing	OSPF 1, router-id 2.2.2.2, area 0, passive on loopback and LAN-facing interfaces	Same design
Traffic control	`SRV4-Inbound` applied **out** on both subinterfaces	Also permits NTP requests to SRV4

3.2.3 Edge routers (Edge-R1 to Edge-R4)

Each Edge router is the single boundary between the OSPF backbone and its region's EIGRP AS:

- **Routing:** the WAN link and loopback are in OSPF area 0, and the site-facing link and loopback are in EIGRP. Loopback0 is passive in both protocols.
- **Redistribution:** both directions, using the commands in Section 2.3.
- **ACLs:** two extended ACLs per router, both applied with `in` (Section 3.5).
- **Unused ports:** G0/2 is shut down.

Router	Region / EIGRP AS	Loopback	Links	ACLs
Edge-R1	A / 1	3.3.3.3	WAN G0/1 `10.10.0.2/30` (HQ-R1); site G0/0 `10.10.0.5/30`	Sales-Inbound, Sales-Outbound
Edge-R2	B / 2	5.5.5.5	WAN G0/1 `10.20.0.2/30` (HQ-R2); site G0/0 `10.20.0.5/30`	Prod-Inbound, Prod-Outbound
Edge-R3	C / 3	7.7.7.7	WAN G0/1 `10.30.0.2/30` (HQ-R1); site G0/0 `10.30.0.5/30`	Finance-Inbound, Finance-Outbound
Edge-R4	D / 4	9.9.9.9	WAN **G0/0** `10.40.0.34/30` (HQ-R2); site **G0/1** `10.40.0.30/30`	Mgmt-Inbound, Mgmt-Outbound

Edge-R4 differs from the others in two ways. Its interface roles are reversed (G0/0 faces the WAN and G0/1 faces the site), and its SSH rules are reversed too, because the Network Engineer is inside Region D.

3.3 Site routers

3.3.1 SiteA-R1, SiteB-R2 and SiteC-R3 (router-on-a-stick sites)

- **Trunk:** G0/0 carries the user, server and management VLANs to the access switch as 802.1Q subinterfaces. G0/1 is the uplink to the Edge router, and G0/2 is shut down.
- **DHCP:** one pool per user VLAN with addresses .1 to .10 excluded. Each pool provides the router as default gateway, DNS server 192.168.6.2 and the domain `yannickccna.com`.
- **Routing:** EIGRP in the region's AS. Loopback0 and every subinterface are passive, so only the uplink forms an adjacency.
- **Filtering:** no ACLs on the site routers. Filtering is done at the Edge.

Router	Region / AS	Loopback	Uplink	Subinterfaces and DHCP pools
SiteA-R1	A / 1	4.4.4.4	G0/1 `10.10.0.6/30`	G0/0.10 `192.168.1.1/24`, G0/0.20 `192.168.2.1/24`, G0/0.99 `10.40.0.41/29`; pools VLAN10, VLAN20
SiteB-R2	B / 2	6.6.6.6	G0/1 `10.20.0.6/30`	G0/0.30 `192.168.3.1/24`, G0/0.40 `192.168.4.1/24`, G0/0.99 `10.40.0.49/29`; pools VLAN30, VLAN40
SiteC-R3	C / 3	8.8.8.8	G0/1 `10.30.0.6/30`	G0/0.50 `192.168.5.1/24`, G0/0.99 `10.40.0.57/29`; pool VLAN50

The first address leased in each pool is .11 (as seen in TK004). The server VLANs 20 and 40 also have pools, but SRV1 and SRV2 use static addresses inside the excluded range.

3.3.2 SiteD-R4 (Region D)

Region D has no router-on-a-stick and no DHCP, since its devices use static addresses. SiteD-R4 is a pure routed hub with three /30 links:

Interface	Address	Neighbor
G0/0	`10.40.0.22/30`	DSW1
G0/1	`10.40.0.26/30`	DSW2
G0/2	`10.40.0.29/30`	Edge-R4

It runs EIGRP AS 4 on those links and on its loopback (11.11.11.11). Edge-R4 performs the redistribution into OSPF.

3.4 Switching configuration

3.4.1 Distribution switches (DSW1 and DSW2)

Function	DSW1	DSW2
Loopback	12.12.12.12	13.13.13.13
Routed uplink to SiteD-R4	G1/0/1 `10.40.0.21/30`	G1/0/1 `10.40.0.25/30`
Link between the DSWs	Port-channel 1 (G1/0/2 to G1/0/3, LACP active), routed, `10.40.0.17/30`	Same, `10.40.0.18/30`
Trunks to ASW4 to ASW6	G1/0/4 to G1/0/6, native VLAN 999, allowed VLANs 60 and 99, `nonegotiate`	Same
Spanning tree	Rapid-PVST. Root for VLAN 99 (priority 0), backup for VLAN 60 (priority 4096)	Root for VLAN 60 (priority 0), backup for VLAN 99 (priority 4096)
SVIs	Vlan60 `10.60.0.2/29`, Vlan99 `10.40.0.2/28`	Vlan60 `10.60.0.3/29`, Vlan99 `10.40.0.3/28`
HSRPv2	VLAN 99 (VIP 10.40.0.1): priority 105 with preempt, **active**. VLAN 60 (VIP 10.60.0.1): default priority, standby	VLAN 60: priority 105 with preempt, **active**. VLAN 99: default priority, standby
Routing	EIGRP AS 4, passive on loopback and SVIs	Same
Unused ports	Shut down	Shut down

The design has two rules. Each switch is the primary gateway for one VLAN, which shares the load between them. The spanning-tree root for each VLAN is the same switch as its HSRP active router, so traffic does not take an extra hop.

3.4.2 Access switches (ASW1 to ASW8)

Switch	Site	Access ports	Uplink and allowed VLANs	Management SVI → gateway	Snooping + DAI
ASW1	A	Fa0/1–2 VLAN 10, Fa0/3 VLAN 20 (SRV1), Fa0/5 VLAN 99	Fa0/4 → SiteA-R1: 10, 20, 99	`10.40.0.43/29` → .41	VLANs 10, 20
ASW2	B	Fa0/1–2 VLAN 30, Fa0/3 VLAN 40 (SRV2), Fa0/5 VLAN 99	Fa0/4 → SiteB-R2: 30, 40, 99	`10.40.0.51/29` → .49	VLANs 30, 40
ASW3	C	Fa0/1–2 VLAN 50, Fa0/4 VLAN 99	Fa0/3 → SiteC-R3: 50, 99	`10.40.0.59/29` → .57	VLAN 50
ASW4	D	Fa0/1 VLAN 99 (Network Engineer)	G0/1 → DSW1, G0/2 → DSW2: 60, 99	`10.40.0.6/28` → 10.40.0.1	None
ASW5	D	Fa0/1 VLAN 99 (IT Manager)	Same	`10.40.0.7/28` → 10.40.0.1	None
ASW6	D	Fa0/1 VLAN 60 (HR)	Same	`10.40.0.8/28` → 10.40.0.1	None
ASW7	HQ	Fa0/1 VLAN 70 (SRV3)	Fa0/2 → HQ-R1: 70, 99	`10.40.0.66/29` → .65	None
ASW8	HQ	Fa0/1 VLAN 80 (SRV4)	Fa0/2 → HQ-R2: 80, 99	`10.40.0.74/29` → .73	None

Common hardening on all eight switches:

- **Spanning tree:** Rapid-PVST with `extend system-id`. STP is disabled for the unused native VLAN 999.
- **Access ports:** PortFast and BPDU guard, `switchport nonegotiate`, port security (sticky MAC, violation `restrict`, one MAC per port), and LLDP and CDP turned off.
- **Trunks:** native VLAN 999, an explicit allowed-VLAN list, and `nonegotiate`.
- **Unused ports and Vlan1:** shut down.
- **DHCP snooping and DAI:** enabled on the user and server VLANs of ASW1 to ASW3 only. The uplink is trusted for both, and option 82 insertion is disabled. Region D and HQ use static addressing, so they need neither.

3.5 ACL policy

The approach is deliberately simple and practical rather than strict. Rules permit what the network needs and deny everything else with a final `deny ip any any`.

Device	ACL	Applied to	Filters
HQ-R1	`SRV3-Inbound`	`out` on G0/2.70 and G0/2.99	Traffic towards SRV3 and the HQ management VLAN
HQ-R2	`SRV4-Inbound`	`out` on G0/0.80 and G0/0.99	Traffic towards SRV4 and the HQ management VLAN
Edge-R1, R2, R3	`Sales-`, `Prod-`, `Finance-Outbound`	`in` on the site-facing G0/0	Traffic leaving the site
Edge-R1, R2, R3	`Sales-`, `Prod-`, `Finance-Inbound`	`in` on the WAN-facing G0/1	Traffic entering the site
Edge-R4	`Mgmt-Outbound` / `Mgmt-Inbound`	`in` on G0/1 (site) / G0/0 (WAN)	Same roles, reversed interfaces
All devices	`SSH-Restriction`	VTY lines	Who may open an SSH session

The names describe traffic relative to the **site**: `*-Outbound` filters traffic leaving the site and `*-Inbound` filters traffic entering it, and both are applied with `in` on the Edge router. Every ACL is built from the same entry types:

- **Routing:** OSPF and EIGRP.
- **Reachability:** ICMP to or from internal subnets, router interfaces and loopbacks only.
- **Services:** DNS and syslog towards SRV3, NTP towards SRV4, and the replies.

- **Management:** SSH from the Network Engineer, with `established` for the replies.
- **Inter-department flows:** Sales ↔ SRV2 and Prod ↔ SRV1 on the Sales and Prod Edge ACLs.

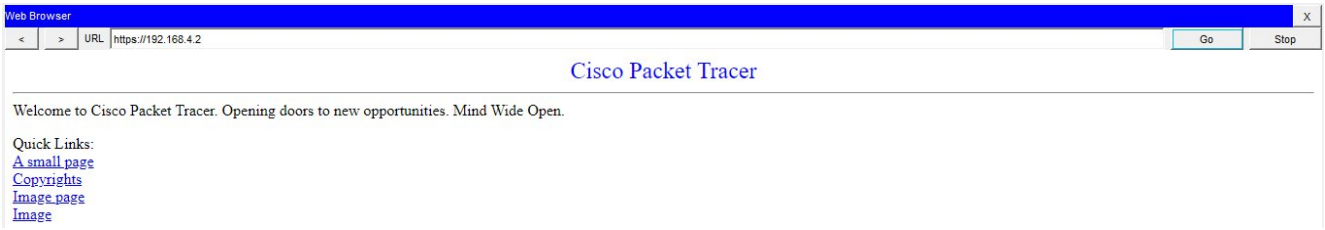
3.5.1 ACL test plan and results

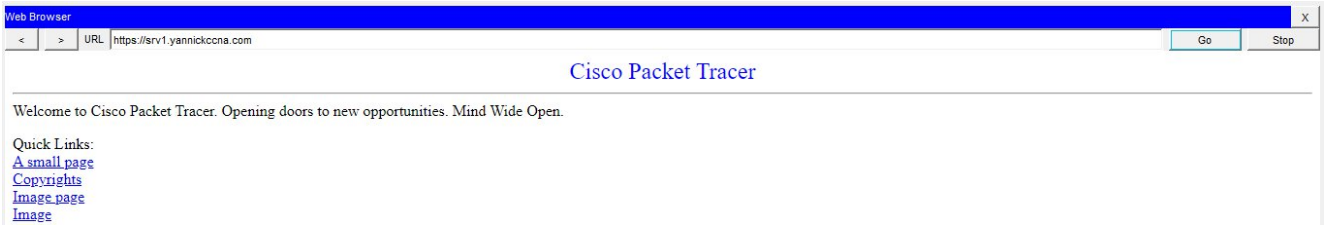
The ACLs were tested once the baseline was in place. Before the tests started, the counters were cleared on every Edge router with `clear access-list counters`, so every counter shown in test 10 comes from these tests only.

#	Test	Expected	Result	Status
1	From the Network Engineer (10.40.0.4), SSH to 10.30.0.5, 5.5.5.5 and ASW1	All log in	All three log in	Pass
2	Technician A to SiteA-R1; Technician B to SiteB-R2; Technician C to 7.7.7.7 (Edge-R3)	All log in	All three log in	Pass
3	Technician A to SiteD-R4; PC3 to Edge-R2; PC6 to HQ-R1 (SSH)	All attempts fail	All three time out	Pass
4	SiteC-R3 ping 10.30.0.5; Edge-R3 ping 10.30.0.6	All pings succeed	5/5 on both	Pass
5	From PC5, ping 10.40.0.34, 10.40.0.30, 10.40.0.29, 9.9.9.9 and 11.11.11.11	All pings succeed	4/4 on each, TTL 250–251	Pass
6	From PC6, ping PC2, PC4, HR, SRV3 and SRV4	All pings succeed	4/4 on each, TTL 121–125	Pass
7	PC1 opens https://192.168.4.2; PC4 and PC5 open https://srv1.yannickccna.com	PC1 and PC4 succeed; PC5 is blocked	PC1 and PC4 open the page; PC5 cannot	Pass
8	`show ntp status` on SiteC-R3	Clock still synchronized	Synchronized, stratum 2, reference 192.168.7.2	Pass
9	`show ip ospf neighbor` and `show ip eigrp neighbors` on Edge-R2	Both adjacencies up	OSPF FULL with 2.2.2.2; EIGRP neighbor 10.20.0.6	Pass
10	`show access-lists` for Sales-Inbound (Edge-R1), Finance-Outbound (Edge-R3) and Mgmt-Inbound (Edge-R4)	Many OSPF matches and no denies on Edge-R1 and Edge-R4; many EIGRP matches and a few denies on Edge-R3; many TCP/UDP matches on all	As expected	Pass

3.5.2 Test evidence

- **Tests 1 and 2 (SSH allowed):** the Network Engineer reaches devices in three regions, and each technician reaches only the devices of their own region. Technician C's session to Edge-R3 is counted by `permit tcp host 10.40.0.58 host 7.7.7.7 eq 22` (23 matches in test 10).
- **Test 3 (SSH refused):** all three attempts end with `% Connection timed out; remote host not responding`. Each source is outside what the destination allows: Region D devices accept the Network Engineer only, Edge-R2 accepts only the Network Engineer and Technician B, and Finance-Outbound on Edge-R3 has no SSH entry for PC6. The 24 deny matches on Edge-R3 in test 10 are consistent with these attempts.
- **Tests 4 to 6 (ICMP allowed):** the pings pass every ACL on their path. The TTL values match the expected path lengths. For example, PC6 reaches SRV3 through three routers (128 − 3 = 125) and SRV4 through four (124), and PC5 reaches SiteD-R4 with TTL 250 and Edge-R4 with TTL 251.
- **Test 7 (web access):** the inter-department entries on Edge-R1 are counted: `permit ip host 192.168.4.2 192.168.1.0 0.0.0.255` (3 matches, PC1 and SRV2) and `permit ip 192.168.3.0 0.0.0.255 host 192.168.2.2` (6 matches, PC4 and SRV1). PC5 is blocked because Finance has no web entry towards SRV1.







- **Test 8 (NTP):** NTP traffic from SiteC-R3 crosses Edge-R3, where `permit udp any host 192.168.7.2 eq 123` shows 649 matches, and the clock stays synchronized.
- **Test 9 (routing):** every Edge ACL starts with `permit ospf any any` and `permit eigrp any any`, so the filters cannot break the adjacencies. The EIGRP uptime column shows 00:00:00, which is the Packet Tracer display quirk noted in Section 5.4.
- **Test 10 (counters):** the matches are as expected. Entries without matches are omitted below.

```
Edge-R1#show access-lists Sales-Inbound
    permit ospf any any (643 match(es))
    permit ip host 192.168.4.2 192.168.1.0 0.0.0.255 (3 match(es))
    permit ip 192.168.3.0 0.0.0.255 host 192.168.2.2 (6 match(es))
    permit udp host 192.168.7.2 eq 123 any (939 match(es))
    permit tcp host 10.40.0.4 any eq 22 (28 match(es))
    deny ip any any                                  <- no matches

Edge-R3#show access-lists Finance-Outbound
    permit eigrp any any (1126 match(es))
    permit udp any host 192.168.7.2 eq 123 (649 match(es))
    permit tcp host 10.40.0.58 host 7.7.7.7 eq 22 (23 match(es))
    deny ip any any (24 match(es))

Edge-R4#show access-lists Mgmt-Inbound
    permit ospf any any (2819 match(es))
    permit udp host 192.168.7.2 eq 123 any (2372 match(es))
    permit tcp any eq 22 host 10.40.0.4 established (80 match(es))
    deny ip any any                                  <- no matches
```

3.6 Baseline verification

This section records the healthy state of the network. Every ticket report in Section 4 compares its evidence against these values, for example the 22-subnet routing table that SiteB-R2 regains in TK003.

3.6.1 Routing adjacencies and table size

Device	OSPF neighbors	EIGRP neighbors	`10.0.0.0/8` entries
HQ-R1	4 (Edge-R1, Edge-R3, HQ-R2 on both serial links)	–	25 subnets, 4 masks
HQ-R2	4 (Edge-R2, Edge-R4, HQ-R1 on both serial links)	–	25 subnets, 4 masks
Edge-R1	1 (HQ-R1)	1 (SiteA-R1)	22 subnets, 4 masks
Edge-R2	1 (HQ-R2)	1 (SiteB-R2)	22 subnets, 4 masks
Edge-R3	1 (HQ-R1)	1 (SiteC-R3)	22 subnets, 4 masks
Edge-R4	1 (HQ-R2)	1 (SiteD-R4)	22 subnets, 4 masks
SiteA-R1, SiteB-R2, SiteC-R3	–	1 each (its Edge router)	22 subnets, 4 masks
SiteD-R4	–	3 (DSW1, DSW2, Edge-R4)	23 subnets, 4 masks
DSW1, DSW2	–	2 each (SiteD-R4, the other DSW over Po1)	24 subnets, 4 masks

All OSPF adjacencies are FULL. Every router also holds the 12 loopbacks and the seven 192.168.x.0/24 networks, so every site reaches every other site. EIGRP neighbor uptime in Packet Tracer shows seconds even when routes are hours old, so route age is the better stability indicator.

3.6.2 Route-type fingerprint

Router type	Routes from its own domain	Routes from the other domain
HQ and Edge routers	`O` (HQ, Edge and HQ server networks)	`O E2 [110/20]`: EIGRP routes redistributed into OSPF
Edge routers (site side)	`D` for their own site	–
Site routers	`D` for the site's own networks	`D EX [170/28416]` for everything else
DSWs	`D` inside Region D	`D EX [170/28672]` for everything else

Examples: on HQ-R2, Site B is `O E2 192.168.3.0/24 [110/20] via 10.20.0.2`. On SiteA-R1, the HQ server network is `D EX 192.168.6.0/24 [170/28416] via 10.10.0.5`.

3.6.3 WAN redundancy and path evidence

- **Primary and backup:** traffic uses the primary S0/3/0 (cost 64, for example `2.2.2.2 [110/65] via 10.0.0.2`), while the backup subnet stays reachable at cost 200 + 1 (`10.0.0.4/30 [110/201]` on the Edge routers).
- **Region D load sharing:** SiteD-R4 has two equal-cost routes to the HSRP networks, one via each DSW.
- **TTL fingerprints** from the tickets confirm the designed paths:

Test	TTL	Path
Region A → SRV2 (TK002, TK003)	122	SiteA-R1, Edge-R1, HQ-R1, HQ-R2, Edge-R2, SiteB-R2
Region B → SRV1 (TK005)	122	The same six routers in reverse
Region D → SRV3 (TK001)	123	DSW1, SiteD-R4, Edge-R4, HQ-R2, HQ-R1

3.6.4 Layer 2 and redundancy baseline (Region D)

Check	DSW1	DSW2
HSRP VLAN 60 (group 1, VIP 10.60.0.1)	Standby, priority 100	**Active**, priority 105, preempt
HSRP VLAN 99 (group 2, VIP 10.40.0.1)	**Active**, priority 105, preempt	Standby, priority 100
Spanning tree VLAN 60	Root port Gi1/0/4 (cost 8); Gi1/0/5 and Gi1/0/6 alternate, blocking	**Root bridge** (priority 0 + 60); all ports designated, forwarding
Spanning tree VLAN 99	**Root bridge** (priority 0 + 99); all ports designated, forwarding	Root port Gi1/0/4 (cost 8); Gi1/0/5 and Gi1/0/6 alternate, blocking
Port-channel 1	`Po1(RU)`, LACP, Gi1/0/2 and Gi1/0/3 bundled `(P)`	Same

- **Alignment:** the HSRP active router and the spanning-tree root are the same switch for each VLAN.
- **Routed Port-channel:** Po1 is Layer 3 (`R`), so the two DSWs share no direct Layer 2 link.
- **HSRP hello path:** the only forwarding Layer 2 path between the DSWs runs through ASW4, since each non-root DSW blocks its ports towards ASW5 and ASW6. HSRP hellos therefore travel DSW1 ↔ ASW4 ↔ DSW2.

3.6.5 Verification commands

Layer	Commands
Routing	`show ip ospf neighbor`, `show ip eigrp neighbors`, `show ip route`, `show ip route <prefix>`
Redundancy	`show standby brief`, `show spanning-tree root`, `show etherchannel summary`
Switching	`show vlan brief`, `show interfaces trunk`, `show port-security`, `show ip dhcp snooping`

4. Incident Response & Ticket Resolution

4.1 Methodology

After the network was built and verified (Section 3.6), five tickets were introduced. Each one describes a problem as a user or operator would report it. The root cause is found by working through the layers from the bottom up and ruling out one layer at a time. When a fault breaks the SSH path, the console is used instead.

Layer	Question	Typical checks
Layer 1	Is the link up?	`show ip interface brief`, `show interfaces status`
Layer 2	Do the VLANs, trunks and switch security features let frames through?	`show vlan brief`, `show interfaces trunk`, `show port-security`, `show ip dhcp snooping`, `show spanning-tree`
Layer 3	Does the packet have a route and a working gateway?	`show ip route`, `show ip ospf neighbor`, `show ip eigrp neighbors`, `show standby`, `show arp`, `ping`
Layer 4	Does an ACL or a service block the traffic?	`show access-lists`, `show ip interface <int> | include access list`, a real service test (web page, SSH)

Every ticket report follows the same template: ticket and symptoms; evidence layer by layer; root cause; resolution; verification. The sections below summarize each ticket. The complete reports with all captures are listed in Appendix D.

4.2 TK001 – HSRP version mismatch

Ticket	**Region D hosts report intermittent loss of connectivity after a disruption on DSW2's upstream path. Duplicate IP warnings appear on the distribution switches.**
Faulty layer	Layer 3 (first-hop redundancy)
Root cause	DSW2 ran HSRP version 1 (the default) while DSW1 ran version 2
Fix	`standby version 2` on DSW2 Vlan60 and Vlan99

- **Symptom:** pings from Region D alternate between replies and "Destination host unreachable" from 10.40.0.3 (DSW2), which gives 50% loss.
- **Syslog:** `%IP-4-DUPADDR` for 10.60.0.1 and 10.40.0.1, sourced by the virtual MACs 0000.0C07.AC01 and 0000.0C07.AC02 (version 1 format).
- **HSRP state:** `show standby brief` shows both switches Active for both VLANs, each with the peer shown as unknown. `show standby` shows DSW1 on version 2 (virtual MAC 0000.0C9F.Fxxx) and DSW2 on version 1.
- **Layers 1, 2 and 4 are clean:** DSW1's uplink and trunks are up, VLANs 60 and 99 are allowed and forwarding, and no ACL touches the SVIs.

Why the loss is intermittent: with two active routers, hosts receive ARP answers with two different virtual MACs and send some packets to DSW1 and some to DSW2. DSW2 had lost its uplink, so it could not forward them. **Why the switches did not see each other:** HSRP versions 1 and 2 use different multicast addresses (224.0.0.2 and 224.0.0.102) and different virtual MAC formats, so each ignores the other's hellos.

```
DSW2(config)#interface vlan 60
DSW2(config-if)#standby version 2
DSW2(config-if)#interface vlan 99
DSW2(config-if)#standby version 2
```

Verification: DSW2 moves to Standby (`%HSRP-6-STATECHANGE … Speak -> Standby`) and a ping from Region D to 192.168.6.2 returns 4/4 replies with TTL 123. **Caveat:** DSW2 has priority 105 with preempt for VLAN 60, so it reclaims the Active role as soon as its uplinks return. If the uplinks stay down, tracking on those interfaces would prevent a dead gateway from being Active.

4.3 TK002 – Router-on-a-stick failure

Ticket	End hosts at every site and at HQ cannot reach SRV2 in Region B, although all other servers are reachable.
Faulty layer	Layer 2
Root cause	VLAN 40 is missing from the allowed-VLAN list of ASW2's uplink trunk (Fa0/4 carried only VLANs 30 and 99)
Fix	`switchport trunk allowed vlan add 40` on ASW2 Fa0/4

- **Symptom:** pings to SRV2 (192.168.4.2) time out from every site, with no ICMP error. The packets are dropped silently or the reply is lost.
- **Layer 1:** ASW2 Fa0/3 (VLAN 40) and the Fa0/4 trunk are connected, and SiteB-R2 G0/0.40 is up/up with 192.168.4.1.
- **Layer 2:** `show interfaces trunk` on ASW2 allows only VLANs 30 and 99. VLAN 40 frames are not carried to the router.
- **Layer 3:** SiteB-R2 has a connected route to 192.168.4.0/24, but a ping from the router to SRV2 fails (0/5) and its ARP table has no entry for SRV2. The local ping also rules out ACLs, so routing is fine and the missing piece is the Layer 2 path.

A router subinterface stays up/up as long as the physical interface is up. It does not check which VLANs the switch actually carries, which is why Layer 1 looked healthy.

```
ASW2(config)#interface f0/4
ASW2(config-if)#switchport trunk allowed vlan add 40
```

Verification: the trunk now allows VLANs 30, 40 and 99. A ping from SiteB-R2 to SRV2 returns 5/5 and a ping from PC1 (Region A) returns 4/4 with TTL 122.

4.4 TK003 – Route redistribution failure

Ticket	Region B users cannot reach any other site or the servers. Internal Region B traffic is fine, and OSPF and EIGRP neighbors show as up.
Faulty layer	Layer 3 (routing)
Root cause	Edge-R2 had no redistribution between OSPF 1 and EIGRP 2
Fix	`redistribute eigrp 2 subnets` under OSPF and `redistribute ospf 1 metric 1000000 100 255 1 1500` under EIGRP

- **Symptom:** from PC3, pings to 192.168.6.2 and 192.168.2.2 return "Destination host unreachable" from 192.168.3.1, the PC's own gateway. A router answers this way when it has no route. The ping to 5.5.5.5 (Edge-R2) works.
- **Layers 1 and 2:** Edge-R2 and SiteB-R2 interfaces are all up/up on a direct link.
- **Layer 3:** both adjacencies are established (OSPF with HQ-R2 and EIGRP with SiteB-R2). SiteB-R2's table holds only its own subnets and 5.5.5.5, and HQ-R2 lacks 192.168.3.0/24 and 192.168.4.0/24. The routing failure is in **both** directions.
- **Root cause:** Edge-R2 is the only boundary router, and its running configuration contains no `redistribute` statement in either process.

```
Edge-R2(config)#router ospf 1
Edge-R2(config-router)#redistribute eigrp 2 subnets
Edge-R2(config-router)#router eigrp 2
Edge-R2(config-router)#redistribute ospf 1 metric 1000000 100 255 1 1500
```

Verification: HQ-R2 shows `O E2 192.168.3.0/24` and `192.168.4.0/24 [110/20] via 10.20.0.2`, SiteB-R2 has `D EX` routes to all other sites (22 subnets with 4 masks), PC1 pings SRV2 with 4/4 replies and TTL 122, and SRV2's web page opens.

4.5 TK004 – DHCP snooping trust

Ticket	Region A (Sales) users cannot communicate with any other device on the network, even within the same region.
Faulty layer	Layer 2 (switch security feature)
Root cause	DHCP snooping on ASW1 did not trust the uplink Fa0/4, so DHCP Offer and Ack messages from SiteA-R1 were dropped
Fix	`ip dhcp snooping trust` on ASW1 Fa0/4

- **Symptom:** PC1 holds the APIPA address 169.254.147.195 with no gateway or DNS server, so it cannot leave its own subnet. That explains the failed pings to SRV1 (a different VLAN) and to SiteA-R1's loopback.
- **Layers 1 and 3:** links are up, and the VLAN10 pool on SiteA-R1 is correct, yet `show ip dhcp binding` is empty. The exchange breaks between the PC and the router.
- **Layer 2:** the VLAN, trunk and port-security checks pass. `show ip dhcp snooping` shows Fa0/4 as untrusted. The PC's Discover goes out normally, but the router's Offer comes back on the untrusted Fa0/4 and the switch discards it.
- **Layer 4:** DHCP never crosses Edge-R1, so its ACL is irrelevant. No ACL exists on SiteA-R1 or ASW1.

```
ASW1(config)#interface f0/4
ASW1(config-if)#ip dhcp snooping trust
```

Verification: Fa0/4 is trusted while Fa0/1 and Fa0/2 stay untrusted. `ipconfig /renew` on PC1 returns 192.168.1.11 with gateway 192.168.1.1 and DNS 192.168.6.2, and SiteA-R1 records the lease against PC1's MAC address.

4.6 TK005 – ACL faults

TK005 combines three complaints that were escalated together. All three are Layer 4 faults, and in each one the interfaces are up and routing is intact.

Issue	Device	Symptom	Root cause	Fix
(a)	Edge-R3	Region C has no connectivity beyond its edge router	`Finance-Inbound` applied **out** on G0/1 instead of in	`no ip access-group Finance-Inbound out`, then `ip access-group Finance-Inbound in`
(b)	Edge-R2	The Network Engineer cannot SSH to Region B devices	`Prod-Outbound` permits TCP **to** port 22, but SSH replies come **from** port 22	`permit tcp any eq 22 host 10.40.0.4 established`
(c)	Edge-R1	SRV1's web pages do not load although ping works	`Sales-Outbound` entry 40 uses `host 192.168.3.0`, which matches a single address	`40 permit ip host 192.168.2.2 192.168.3.0 0.0.0.255`

- **(a) Direction:** a Region C PC pings Edge-R3's loopback successfully, but a ping to 192.168.6.2 returns "unreachable" from 10.30.0.5, Edge-R3's own site-facing address. Edge-R3 has a full routing table, so it is refusing the packet because of an ACL. `show ip interface g0/1 | include access list` showed `Finance-Inbound` as the outgoing ACL.
- **(b) Port position:** `Prod-Outbound` is applied inbound on Edge-R2's site-facing G0/0, so it filters everything that leaves Region B. The SSH request passes `Prod-Inbound` and reaches SiteB-R2, but the reply has source port 22 and is stopped. After the fix, `ssh -l yannick 6.6.6.6` opens the SiteB-R2 prompt.
- **(c) Wildcard:** `Sales-Outbound` is applied inbound on Edge-R1's site-facing G0/0. The replies from SRV1 to Region B hosts did not match entry 40 and were dropped by the final deny, so the web page failed while ICMP, matched by another entry, worked. After the fix, entry 40 shows match counters and the page loads.

4.7 Summary matrix

Ticket	Symptom	Faulty layer	Device	Root cause	Key command
TK001	Intermittent loss, duplicate IP warnings	3	DSW2	HSRP version 1 against version 2	`show standby`
TK002	SRV2 unreachable from all sites	2	ASW2	VLAN 40 missing from the trunk allowed list	`show interfaces trunk`
TK003	Region B isolated, neighbors up	3	Edge-R2	No redistribution between OSPF and EIGRP	`show running-config`, route tables on both sides
TK004	APIPA address in Region A	2	ASW1	DHCP snooping did not trust the uplink	`show ip dhcp snooping`
TK005	Three ACL-related failures	4	Edge-R3, R2, R1	Wrong direction, wrong port position, wrong address match	`show ip interface … | include access list`, `show access-lists`

5. Key Takeaways & Lessons Learned

5.1 Troubleshooting methodology

- **Work from the bottom up and rule layers out quickly.** In TK003, one `show ip interface brief` on each router closed Layers 1 and 2 and left routing as the next suspect.
- **Read who sends the error.** "Destination host unreachable" from the local gateway means a missing route (TK003). The same message from a router that has a complete routing table means an ACL (TK005a). Plain timeouts point to filtering, a missing return path or a Layer 2 gap (TK002).
- **Compare against the baseline.** Route counts, HSRP roles and spanning-tree roots from Section 3.6 turn "something is wrong" into "this is missing".
- **Check both sides of a boundary.** A redistribution fault shows up as missing routes on both sides of the boundary router (TK003).
- **Use the console when the fault breaks the path.** SSH depends on routing and ACLs, so the console is the reliable fallback.

5.2 Recurring fault patterns

Pattern	Tickets	Why it misleads	What to check
Status is healthy but the service is broken	TK001, TK002, TK003	An up/up subinterface, a FULL neighbor or a connected route does not prove forwarding or route exchange	Test end to end and compare route tables on both sides
One device of a pair or link is configured differently	TK001, TK002	Each device looks correct on its own	Compare `show standby` and `show interfaces trunk` on both ends
A security feature drops traffic silently	TK004	Snooping, DAI and port security leave no obvious error	`show ip dhcp snooping`, trusted ports, `show port-security`
An ACL does not act as it reads	TK005	Direction, port position and wildcard errors look like valid syntax	`show ip interface`, `show access-lists` with counters

5.3 Preventive practices

- Keep a saved baseline (Section 3.6) and compare after every change.
- Verify each change with a ping and the matching `show` command before closing it.
- Configure redundancy features as a pair and compare both ends: HSRP version, priority and preempt, and the trunk allowed lists.
- Name ACLs by a single convention and document the direction rule (Section 3.5). Use explicit sequence numbers so that one entry can be replaced without touching the rest.
- Add uplink tracking to HSRP so that a switch without an upstream path does not stay Active.
- Trust only the ports that lead to the DHCP server, and test a DHCP request after enabling snooping.
- After any redistribution change, compare route counts with the baseline.

5.4 Lab and tool observations

- **Command support:** `logging trap informational` is not accepted in Packet Tracer, so the trap level is `debugging`.
- **EIGRP uptime:** neighbor uptime shows seconds while routes are hours old. Use route age to judge stability.
- **Stale entries:** captures taken after a reconfiguration can contain leftovers (TK003 showed a `D EX 10.20.0.0/30` entry before redistribution was configured). Clear the routing processes or reload before taking "before" captures.
- **DAI and static hosts:** no impact was seen on SRV1 and SRV2, which use static addresses. On real IOS, DAI checks ARP against the snooping binding table, so such hosts would need ARP ACL entries or static bindings.

6. Conclusion & Future Enhancements

6.1 Conclusion

The project delivered a complete multi-site network with documented design, configuration and baseline, and five incidents that were diagnosed and resolved. The network combines OSPF, EIGRP and redistribution, HSRPv2 redundancy, router-on-a-stick, DHCP with layered switch security, and ACL-based traffic control on every Edge and HQ router.

The incidents show that failures in a network of this kind seldom appear as a down link. They appear as a missing VLAN on a trunk, a version mismatch between two switches, a missing redistribution statement, a trust setting or an ACL that does not act as written. A layered method, a documented baseline and a consistent report template made each one fast to isolate.

6.2 Project limitations

- **Simulation scope:** Packet Tracer does not support every IOS feature and displays some counters differently (Section 5.4).
- **Permissive ACLs by design:** the ACLs favor simplicity over strictness. For example, ICMP is allowed broadly between internal addresses.
- **Management access:** the console has no login and the VTY lines have no idle timeout. Authentication is local only, with no AAA server.
- **Single WAN path per region:** only HQ has a redundant link. A failed Edge router isolates its region.
- **No HSRP tracking:** a distribution switch that loses its uplink stays Active for its VLAN (the TK001 situation).
- **Static hosts with DAI:** servers on DAI-protected VLANs would need ARP ACL entries on real equipment.
- **Sticky port security:** learned MAC addresses are stored in the configuration, so replacing a device requires clearing the entry.

6.3 Future enhancements

Enhancement	Benefit
Centralized AAA (TACACS+ or RADIUS) with per-role privilege levels	Individual accounts, audit trail and no shared local credentials
HSRP interface tracking on the DSW uplinks	Automatic failover when an upstream path is lost (TK001)
Route filtering or tagging on redistribution	Loop protection if a region gets a second boundary router
OSPF and EIGRP authentication	Protection against rogue routers on the WAN links
A second WAN path or Edge router per region	Removes the single point of failure in each region
Tighter ACLs (specific ICMP types, per-service entries)	Narrower exposure while keeping the same structure
ARP ACLs or static bindings for DAI on server ports	Correct DAI behavior on real equipment
SNMP and syslog severity tuning with a monitoring server	Earlier detection of faults like those in the five tickets
IPv6 dual-stack	Preparation for IPv6 services

7. Appendices

Appendix A – Inventory, links and addressing

A.1 Hardware inventory

Device type	Model	Quantity
Routers	Cisco 2911	10
Distribution switches	Catalyst 3650 24PS	2
Access switches	Catalyst 2960	8

A.2 Infrastructure links

Group	Endpoint A	Endpoint B	Note
HQ core	HQ-R1 S0/3/0	HQ-R2 S0/3/0	Primary serial link
HQ core	HQ-R1 S0/3/1	HQ-R2 S0/3/1	Backup serial link (cost 200)
HQ to Edge	HQ-R1 G0/0	Edge-R1 G0/1	
HQ to Edge	HQ-R1 G0/1	Edge-R3 G0/1	
HQ to Edge	HQ-R2 G0/1	Edge-R2 G0/1	
HQ to Edge	HQ-R2 G0/2	Edge-R4 G0/0	
HQ LAN	HQ-R1 G0/2	ASW7 F0/2	Trunk, VLANs 70 and 99
HQ LAN	HQ-R2 G0/0	ASW8 F0/2	Trunk, VLANs 80 and 99
Edge to site	Edge-R1 G0/0	SiteA-R1 G0/1	
Edge to site	Edge-R2 G0/0	SiteB-R2 G0/1	
Edge to site	Edge-R3 G0/0	SiteC-R3 G0/1	
Edge to site	Edge-R4 G0/1	SiteD-R4 G0/2	
Site LAN	SiteA-R1 G0/0	ASW1 F0/4	Trunk, VLANs 10, 20, 99
Site LAN	SiteB-R2 G0/0	ASW2 F0/4	Trunk, VLANs 30, 40, 99
Site LAN	SiteC-R3 G0/0	ASW3 F0/3	Trunk, VLANs 50, 99
Region D	SiteD-R4 G0/0	DSW1 G1/0/1	Routed
Region D	SiteD-R4 G0/1	DSW2 G1/0/1	Routed
Region D	DSW1 G1/0/2–3	DSW2 G1/0/2–3	Port-channel 1 (LACP), routed
Region D	DSW1 G1/0/4	ASW4 G0/1	Trunk, VLANs 60 and 99
Region D	DSW2 G1/0/4	ASW4 G0/2	Trunk, VLANs 60 and 99
Region D	DSW1 G1/0/5	ASW5 G0/1	Trunk, VLANs 60 and 99
Region D	DSW2 G1/0/5	ASW5 G0/2	Trunk, VLANs 60 and 99
Region D	DSW1 G1/0/6	ASW6 G0/1	Trunk, VLANs 60 and 99
Region D	DSW2 G1/0/6	ASW6 G0/2	Trunk, VLANs 60 and 99

A.3 End-device connections

Switch	Port	Device	VLAN
ASW1	Fa0/1, Fa0/2	PC1, PC2 (Sales)	10
ASW1	Fa0/3	SRV1	20
ASW1	Fa0/5	Technician A	99
ASW2	Fa0/1, Fa0/2	PC3, PC4 (Prod)	30

Switch	Port	Device	VLAN
ASW2	Fa0/3	SRV2	40
ASW2	Fa0/5	Technician B	99
ASW3	Fa0/1, Fa0/2	PC5, PC6 (Finance)	50
ASW3	Fa0/4	Technician C	99
ASW4	Fa0/1	Network Engineer	99
ASW5	Fa0/1	IT Manager	99
ASW6	Fa0/1	HR Manager	60
ASW7	Fa0/1	SRV3	70
ASW8	Fa0/1	SRV4	80

A.4 Point-to-point subnets

Link	Subnet	Endpoint A	Endpoint B
HQ-R1 – HQ-R2 (primary)	10.0.0.0/30	HQ-R1 S0/3/0 .1	HQ-R2 S0/3/0 .2
HQ-R1 – HQ-R2 (backup)	10.0.0.4/30	HQ-R1 S0/3/1 .5	HQ-R2 S0/3/1 .6
HQ-R1 – Edge-R1	10.10.0.0/30	HQ-R1 G0/0 .1	Edge-R1 G0/1 .2
HQ-R1 – Edge-R3	10.30.0.0/30	HQ-R1 G0/1 .1	Edge-R3 G0/1 .2
HQ-R2 – Edge-R2	10.20.0.0/30	HQ-R2 G0/1 .1	Edge-R2 G0/1 .2
HQ-R2 – Edge-R4	10.40.0.32/30	HQ-R2 G0/2 .33	Edge-R4 G0/0 .34
Edge-R1 – SiteA-R1	10.10.0.4/30	Edge-R1 G0/0 .5	SiteA-R1 G0/1 .6
Edge-R2 – SiteB-R2	10.20.0.4/30	Edge-R2 G0/0 .5	SiteB-R2 G0/1 .6
Edge-R3 – SiteC-R3	10.30.0.4/30	Edge-R3 G0/0 .5	SiteC-R3 G0/1 .6
Edge-R4 – SiteD-R4	10.40.0.28/30	Edge-R4 G0/1 .30	SiteD-R4 G0/2 .29
SiteD-R4 – DSW1	10.40.0.20/30	SiteD-R4 G0/0 .22	DSW1 G1/0/1 .21
SiteD-R4 – DSW2	10.40.0.24/30	SiteD-R4 G0/1 .26	DSW2 G1/0/1 .25
DSW1 – DSW2	10.40.0.16/30	DSW1 Po1 .17	DSW2 Po1 .18

A.5 Management VLAN 99

Site	Subnet	Addresses
HQ (HQ-R1)	10.40.0.64/29	HQ-R1 G0/2.99 .65, ASW7 .66
HQ (HQ-R2)	10.40.0.72/29	HQ-R2 G0/0.99 .73, ASW8 .74
Region A	10.40.0.40/29	SiteA-R1 G0/0.99 .41, Technician A .42, ASW1 .43
Region B	10.40.0.48/29	SiteB-R2 G0/0.99 .49, Technician B .50, ASW2 .51
Region C	10.40.0.56/29	SiteC-R3 G0/0.99 .57, Technician C .58, ASW3 .59
Region D	10.40.0.0/28	HSRP VIP .1, DSW1 .2, DSW2 .3, Network Engineer .4, IT Manager .5, ASW4 .6, ASW5 .7, ASW6 .8

A.6 User and server networks

VLAN	Subnet	Gateway and hosts
Region A – VLAN 10 (Sales)	192.168.1.0/24	SiteA-R1 G0/0.10 .1; PCs by DHCP
Region A – VLAN 20 (SRV1)	192.168.2.0/24	SiteA-R1 G0/0.20 .1; SRV1 .2
Region B – VLAN 30 (Prod)	192.168.3.0/24	SiteB-R2 G0/0.30 .1; PCs by DHCP
Region B – VLAN 40 (SRV2)	192.168.4.0/24	SiteB-R2 G0/0.40 .1; SRV2 .2

VLAN	Subnet	Gateway and hosts
Region C – VLAN 50 (Finance)	192.168.5.0/24	SiteC-R3 G0/0.50 .1; PCs by DHCP
Region D – VLAN 60 (HR)	10.60.0.0/29	HSRP VIP .1, DSW1 .2, DSW2 .3, HR Manager .4
HQ – VLAN 70 (SRV3)	192.168.6.0/24	HQ-R1 G0/2.70 .1; SRV3 .2
HQ – VLAN 80 (SRV4)	192.168.7.0/24	HQ-R2 G0/0.80 .1; SRV4 .2

A.7 Loopbacks and routing membership

Device	Loopback0	Device	Loopback0	Routing protocol
HQ-R1	1.1.1.1	Edge-R3	7.7.7.7	OSPF area 0: HQ-R1, HQ-R2, Edge-R1 to Edge-R4
HQ-R2	2.2.2.2	SiteC-R3	8.8.8.8	EIGRP AS 1: Edge-R1, SiteA-R1
Edge-R1	3.3.3.3	Edge-R4	9.9.9.9	EIGRP AS 2: Edge-R2, SiteB-R2
SiteA-R1	4.4.4.4	SiteD-R4	11.11.11.11	EIGRP AS 3: Edge-R3, SiteC-R3
Edge-R2	5.5.5.5	DSW1	12.12.12.12	EIGRP AS 4: Edge-R4, SiteD-R4, DSW1, DSW2
SiteB-R2	6.6.6.6	DSW2	13.13.13.13	

Appendix B – Exploring the lab

The full running configurations are not reproduced here. The Packet Tracer file contains all 20 devices, and any configuration can be read directly with `show running-config`.

Item	Details
Lab file	[file name and download link]
Software	Cisco Packet Tracer [version]
Console access	No login required
Privileged mode	`enable`, password `ccna`
SSH access	User `yannick`, password `cisco`. Only the sources in the SSH policy (Section 3.1) are accepted, so connect from the Network Engineer PC (10.40.0.4) or use the console
Text configs (optional)	[link to the 20 configuration files, if published]

Suggested way to explore

- **Start with the topology** (Figure 2.1) and the addressing tables in Section 2 and Appendix A.
- **Confirm the baseline** with the commands below and compare the results with Section 3.6.
- **Read the ticket summaries** in Section 4.1 and the ACL tests in Section 3.5, and repeat any test from the PCs.

Device	Command	What to expect
HQ-R2	`show ip route`	25 subnets with 4 masks in the 10.0.0.0/8 block; `O E2` routes for the regions
SiteB-R2	`show ip route`	22 subnets with 4 masks; `D EX` routes for everything outside Region B
DSW1, DSW2	`show standby brief`	DSW1 Active for VLAN 99 and DSW2 Active for VLAN 60, each Standby for the other
Edge-R3	`show access-lists`	Match counters on the Finance ACLs, including the deny entry
ASW1	`show ip dhcp snooping`	Fa0/4 trusted; Fa0/1 and Fa0/2 untrusted

Appendix C – Command quick reference

Layer	Command	Use
1	`show ip interface brief`	Interface status and addresses on routers
1	`show interfaces status`	Port status, VLAN and trunk on switches
2	`show vlan brief`	VLANs defined and ports assigned
2	`show interfaces trunk`	Trunk status, native VLAN and allowed VLANs
2	`show port-security interface <int>`	Port security state and violations
2	`show ip dhcp snooping`	Snooping VLANs and trusted ports
2	`show spanning-tree vlan <n>`	Root bridge and port roles
2	`show etherchannel summary`	Port-channel and LACP members
3	`show ip route [prefix]`	Routing table and route sources
3	`show ip ospf neighbor`, `show ip eigrp neighbors`	Routing adjacencies
3	`show standby [brief]`	HSRP version, state, priority and virtual MAC
3	`show arp`, `show ip dhcp binding`	ARP resolution and DHCP leases
4	`show access-lists`	ACL entries and match counters
4	`show ip interface <int> | include access list`	Which ACL is applied, and in which direction
4	`show running-config | section router`	Routing configuration, including redistribution

Reading TTL: a reply TTL of 128 minus the number of routers crossed identifies the path (for example 122 after six routers). A different value means a different path.

Appendix D – Individual ticket reports

The full reports, each with all captures and the verification, are provided as separate Word documents:

Ticket	File	Subject
TK001	TK001_HSRP_Version_Mismatch_Report.docx	HSRP version mismatch
TK002	TK002_ROAS_Failure_Report.docx	Router-on-a-stick failure
TK003	TK003_Route_Redistribution_Report.docx	Route redistribution failure
TK004	TK004_DHCP_Snooping_Report.docx	DHCP snooping trust
TK005	TK005_ACL_Faults_Report.docx	Three ACL faults